MTH 200 4.3 Minimums & Maximums

Directions: The goal of the following problems is to help challenge and deepen your understanding of the current topic. Don't assume that you will know how to answer each and every question. You will work with at least two other students in the class to try and find solutions and write arguments for each of the following questions. Be sure that you question your classmates' reasoning. If you don't understand how they are getting something you should ASK THEM TO EXPLAIN! Being able to defend your reasoning is important and so you are actually helping them by asking them to explain.

Challenge Questions

1. Determine all critical points of $f(x)=\sqrt{1-x^2}$.

2. Consider the function $f(x)=\dfrac{1}{\sqrt{x-2}}$

a) Confirm that $x=2$ either makes $f'(x)$ equal 0 or become undefined.

b) Explain why $x=2$ is *NOT* a critical point of $f(x)$.

3. Find all critical points of $f(x)=\dfrac{x^2}{x^3+8}$.

EXTENSIONS

a) $y = x^2 (x-1)^{2/3}$

b) $f(x) = \sin|x|$.

SUMMARY POINT: We need to understand what we need to look for when finding critical values. We are looking for

(1) A value of x that makes $f'(x) = 0$ or $f'(x)$ is undefined.

(2) A value of x that is in the domain of $f(x)$.

4. Consider the function $f(x)=\dfrac{ax+b}{x^2-1}$ where a and b are real numbers. Find values of a and b such that $f(x)$ has a local extreme value of 1 at $x=3$.

5. TRUE or FALSE: A function $f(x)$ must have an absolute maximum or minimum on $[a,b]$.

EXTENSION:

6. TRUE or FALSE: The value $y=-10$ cannot be the absolute maximum of a function $f(x)$.

<u>SUMMARY POINT:</u> Remember that the Extreme Value Theorem (EVT) tells us that if we have a continuous function on $[a,b]$ then this function must obtain an absolute maximum and absolute minimum on this interval. These extremes could either happen at a critical point within the interval or at the endpoints of the interval.

7. Find the absolute maximum and absolute minimum values of $f(x)=x^3-3x^2-9x+1$ on $[-2,1]$.

8. One way of proving that $f(x) \le g(x)$ for all x in a given interval is to show that $0 \le g(x) - f(x)$ for all x in the interval; and one way of proving the later inequality is to show that the absolute minimum of $g(x) - f(x)$ on the interval is non-negative. Use this idea to prove that $\sin(x) \le x$ for all x in the interval $[0, 2\pi]$.

9. The concentration $C(t)$ of a drug in the bloodstream t hours after it has been injection is commonly model by the equation $C(t) = \dfrac{K\left(e^{-bt} - e^{-at}\right)}{a-b}$, where $K > 0$ & $a > b > 0$. At what time does the maximum concentration occur?